# Unit 2 Guided Notes — Exponent Rules

Name _____ Date _____ Period _____

Same base? Move the exponents.

**🎯 By the end of this unit, I can…**

- Apply the product, quotient, and power rules of exponents
- Simplify expressions with negative and zero exponents

*Standards: NC.8.EE.1*

Fill in the blanks as we go. For every example and problem, **write each step on its own line** in the space given, and box your final answer.

## 1 Exponent Rules

**📖 Words to know**

**Base**: the number being multiplied. In $2^5$, the base is 2.

**Exponent**: the small number up high. It tells how many times to multiply the base. In $2^5$, the exponent is 5.

**Coefficient**: the number in front of a variable. In $4x^3$, the coefficient is 4.

**Monomial**: one term: a number, a variable, or a product of them, like $4x^3$ or $-2xy^2$.

**📌 KEY RULE:** Exponent rules only apply when the BASES are the SAME!

### Product Rule — Count the Factors

$$x^3 \quad \boxed{x}\boxed{x}\boxed{x}$$
$$\times\, x^2 \quad \boxed{x}\boxed{x}$$
$$= x^5$$

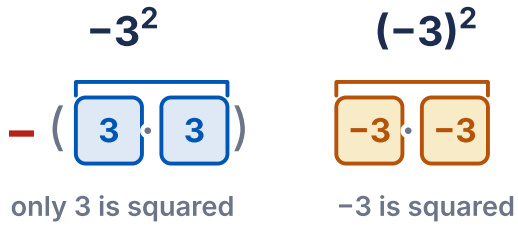
3 + 2 = 5 factors of x

$x^3 \times x^2 = x^{3+2} = x^5$. Count the factors: 3 + 2 = 5 → **ADD** the exponents.

## The Five Core Rules

| Rule | Formula | Simple Example | Monomial Example | Key Phrase |
|---|---|---|---|---|
| Product Rule | $x^a \times x^b = x^{a+b}$ | _____ | _____ | _____ |
| Quotient Rule | $x^a \div x^b = x^{a-b}$ | _____ | _____ | _____ |
| Power Rule | $(x^a)^b = x^{a \cdot b}$ (multiply the exponents) | _____ | _____ | _____ |
| Zero Exponent | $x^0 = 1$ | _____ | _____ | _____ |
| Negative Exponent | $x^{-n} = \frac{1}{x^n}$ | _____ | _____ | _____ |

⚠️ **PARENTHESES MATTER!** $-3^2$ = _____ vs. $(-3)^2$ = _____. Without ( ), ONLY the base gets the exponent — not the negative sign!




The little $^2$ belongs to whatever sits right next to it: the 3 alone, or the whole (−3).

## Modeling the Rules — See WHY

*Each rule comes from writing out the repeated multiplication.*

**Product Rule · $x^3 \times x^2 = x^5$**

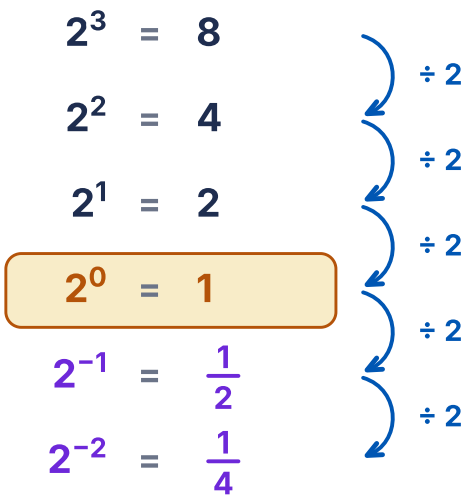
### Quotient Rule · $x^5 \div x^2 = x^3$

### Power Rule · $(x^2)^3 = x^6$

### Zero Exponent · $x^0 = 1$

### Negative Exponent · $x^{-3} = \frac{1}{x^3}$

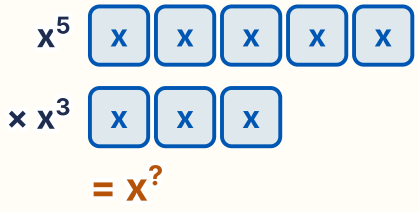
## The Pattern Behind $x^0$ and $x^{-n}$



Each step down the exponent drops by 1 and the value is **÷ 2**. Keep going past $2^1$ and you get $2^0 = 1$, then fractions.

## Quick Check

**MC** **Simplify $x^5 \cdot x^3$. Which rule applies, and what is the result?**




Same base, multiplied: count every x tile.

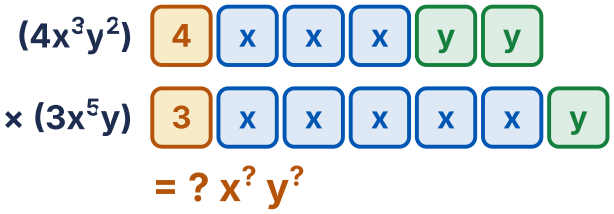
- ○ $x^8$ (Product Rule)
- ○ $x^{15}$ (Power Rule)
- ○ $x^2$ (Quotient Rule)

## Match the Rule to its Key Phrase

**MATCH** **Write the letter of each rule's key phrase on the line.**

| | |
|---|---|
| _____ 1. Product Rule | **A.** Power to power? MULTIPLY |
| _____ 2. Quotient Rule | **B.** Same base? ADD exponents |
| _____ 3. Power Rule | **C.** Flip to a fraction |
| _____ 4. Negative Exponent | **D.** Same base? SUBTRACT exponents |

## Worked Example — Simplify $(4x^3y^2)(3x^5y)$




Multiply the gold numbers. Count the blue x tiles and the green y tiles.

**Step 1 — Multiply coefficients:** $4 \times 3 = 12$

**Step 2 — Add x exponents:** $x^3 \cdot x^5 = x^8$

**Step 3 — Add y exponents:** $y^2 \cdot y^1 = y^3$

✓ **Step 4 — Combine:** $12x^8y^3$

**ERROR ANALYSIS** Simplify $(3x^2)^3$. A student wrote $3x^6$. What went wrong?

$(3x^2)$ [3] [x] [x]

$\times (3x^2)$ [3] [x] [x]

$\times (3x^2)$ [3] [x] [x]

$(3x^2)^3$ means three copies of $3x^2$. Look at the gold tiles: how many 3s are there?

## Part 2 — Practice Problems

📌 **Show every step on its own line.** Box your final answer.

### A. Exponent Rules with Monomials

**1) Simplify $x^3 \cdot x^5$**

$x^3$ [x] [x] [x]

$\times x^5$ [x] [x] [x] [x] [x]

$= x^{?}$

Count all the x tiles.

**Answer:** _____

**2) Simplify $(y^4)^3$**

$y^4$ [y] [y] [y] [y]

$\times y^4$ [y] [y] [y] [y]

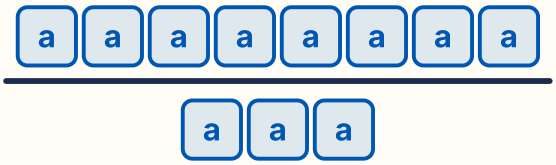
$\times y^4$ [y] [y] [y] [y]

$= y^{?}$

$(y^4)^3$ means 3 copies of $y^4$.

**Answer:** _____

### 3) Simplify $a^8 \div a^3$

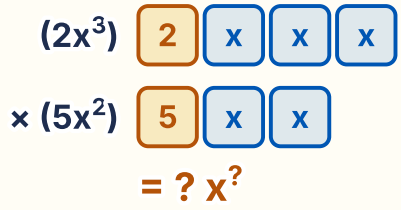



$= a^{?}$

Cancel matching pairs, top and bottom.

**Answer:** _____

### 4) Simplify $(2x^3)(5x^2)$

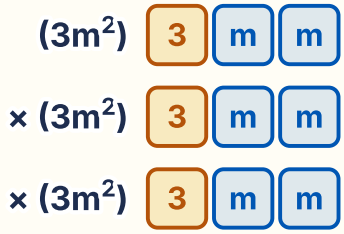



$= ?\ x^{?}$

Multiply the numbers; count the x tiles.

**Answer:** _____

### 5) Simplify $(3m^2)^3$

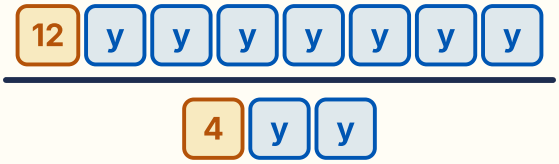



$= ?\ m^{?}$

3 copies: every copy has a 3 and two m tiles.

**Answer:** _____

### 6) Simplify $12y^7 \div 4y^2$




$= ?\ y^{?}$

Divide the numbers; cancel matching y pairs.

**Answer:** _____

**7) Simplify $(-2a^4)^2$**

$(-2a^4)$ [−2] [a] [a] [a] [a]

$\times (-2a^4)$ [−2] [a] [a] [a] [a]

$= ?\ a^?$

2 copies: (−2)(−2) and all the a tiles.

**Answer:** _____

**8) Simplify $(5x^2y)(3xy^3)$**

$(5x^2y)$ [5] [x] [x] [y]

$\times (3xy^3)$ [3] [x] [y] [y] [y]

$= ?\ x^?\ y^?$

Multiply the numbers; count x tiles and y tiles.

**Answer:** _____

**9) Simplify $18n^6 \div 6n^6$**

$$\frac{[18]\,[n]\,[n]\,[n]\,[n]\,[n]\,[n]}{[6]\,[n]\,[n]\,[n]\,[n]\,[n]\,[n]}$$

$= ?$

Divide the numbers; cancel matching n pairs.

**Answer:** _____

**10) Simplify $(4b^3)^2 \div 2b$**

$$\frac{([4]\,[b]\,[b]\,[b])\,([4]\,[b]\,[b]\,[b])}{[2]\,[b]}$$

$= ?\ b^?$

$(4b^3)^2$ is 2 copies of $4b^3$ on top.

**Answer:** _____

**11) Simplify $(-3x^2)(-4x^3)$**

$(-3x^2)$ [−3] [x] [x]

$\times (-4x^3)$ [−4] [x] [x] [x]

$= ? \; x^{?}$

Watch the signs: negative × negative.

**Answer:** _____

**12) Simplify $(x^5y^2)^3$**

$(x^5y^2)$ [x] [x] [x] [x] [x] [y] [y]

$\times (x^5y^2)$ [x] [x] [x] [x] [x] [y] [y]

$\times (x^5y^2)$ [x] [x] [x] [x] [x] [y] [y]

$= x^{?} \; y^{?}$

3 copies of $x^5y^2$.

**Answer:** _____